WEEK 5 TASK

Drafting a Clinical Study Report Section

Detailed Methodology • Statistical Analysis • Discussion of Results

Clinical context: Illustrative Phase II randomised controlled trial of an investigational oral therapy added to stable metformin in adults with type 2 diabetes mellitus.

Prepared as an academic model report

Important academic integrity note: This is a model report using a hypothetical study and illustrative results. It must not be represented as findings from a real clinical trial. Replace the study details and numerical results with the assigned study's verified information if your internship task specifies a particular trial.

1. Detailed Methodology

1.1 Study overview and rationale

This section presents a model clinical study report for a hypothetical Phase II trial evaluating an investigational oral therapy (“Study Drug X”) in adults with type 2 diabetes mellitus whose glycaemic control remains inadequate despite stable background metformin therapy. The example is designed to demonstrate transparent reporting of study methods and interpretation; it is not a report of a real trial. Any numerical results below are illustrative and must be replaced with verified study data before the document is used as a factual report.

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1.2 Study design

The hypothetical study used a Phase II, multicentre, randomised, double-blind, parallel-group, placebo-controlled design. Eligible participants were allocated in a 1:1 ratio to Study Drug X or matching placebo, with both groups continuing stable metformin. Randomisation was selected to reduce systematic differences between treatment groups. Blinding of participants, investigators, study staff involved in assessments, and the sponsor’s blinded study team was intended to limit expectation and assessment bias. A placebo control was appropriate because participants continued standard background treatment and because the purpose was to estimate the incremental effect of Study Drug X.

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1.3 Study population and participant selection

The target population was adults aged 18–75 years with documented type 2 diabetes, HbA1c within a prespecified range of 7.0% to 10.0% at screening, and a stable metformin dose for at least eight weeks before screening. Participants needed to be able to understand the study information, provide written informed consent, and comply with visits and study procedures. These criteria were intended to recruit participants likely to benefit from improved glycaemic control while reducing avoidable clinical variability.

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1.4 Randomisation, allocation concealment, and blinding

An independent statistician would generate the randomisation schedule using validated software and permuted blocks of varying undisclosed sizes, stratified by study site and baseline HbA1c category. The allocation sequence would be held by an authorised system or pharmacy function separate from outcome assessment. Participants would be assigned through a secure interactive response system after eligibility confirmation. Identical packaging and appearance of Study Drug X and placebo would preserve blinding as far as practicable.

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1.5 Study treatment, adherence, and concomitant care

Participants would receive the assigned study treatment according to the protocol and continue their established metformin regimen unless a medically necessary change was required. Dose interruptions, reductions, permanent discontinuations, missed doses, and changes in background medication would be recorded with dates and reasons. Adherence would be assessed through participant diaries, returned medication counts, and questioning at visits. These measures are complementary and may not perfectly capture actual medicine-taking behaviour.

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1.6 Outcomes and assessment procedures

The primary endpoint was change in HbA1c from baseline to Week 24, calculated as Week 24 value minus baseline value. Blood samples would be analysed by a qualified central laboratory using a validated assay with documented quality-control procedures. Sample collection, processing, storage, shipment, and analysis would follow a laboratory manual. Where possible, the same laboratory and consistent procedures would be used across participants to reduce measurement variability.

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1.7 Safety oversight and risk management

Risk management included eligibility screening, review of medical history and concomitant medicines, scheduled laboratory monitoring, clear instructions for reporting symptoms, and access to qualified clinical care. Investigators would assess adverse-event seriousness, severity, outcome, and relationship to study treatment using prespecified definitions. Clinically significant abnormalities would be followed until resolution, stabilisation, or an appropriate clinical endpoint. Pregnancy, suspected overdose, medication errors, and other special situations would be managed and reported according to the protocol and regulatory requirements.

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1.8 Data quality and protocol deviations

Electronic case report forms would be completed from source documents by trained staff. Range checks, logical consistency checks, and review of missing or implausible values would be conducted before database lock. Queries would be resolved against source records without changing the original record invisibly. Important protocol deviations—such as incorrect eligibility, prohibited medication use, or missed primary endpoint assessments—would be classified before unblinding where feasible. The report would describe deviations by treatment group and assess whether they could affect interpretation. A participant flow diagram would show numbers screened, randomised, treated, completing follow-up, and included in each analysis population.

2. Statistical Analysis

2.1 General principles and analysis populations

The SAP would be approved before database lock and unblinding. All statistical tests would be two-sided, with a nominal significance level of 0.05 for the primary endpoint. Effect estimates would be reported with 95% confidence intervals (CIs), not only p-values, because the size and precision of the effect are important for clinical interpretation. Analyses would be performed using validated statistical software, with program specifications, output review, and reproducible records retained.

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2.2 Sample size and power

For planning, assume a between-group difference of 0.5 percentage points in mean HbA1c change, a common standard deviation of 1.0 percentage point, 90% power, and a two-sided 5% significance level. Under a simple two-independent-group approximation, approximately 85 participants per group would be required; allowing for about 15% attrition would increase the target to roughly 100 participants per group, or 200 in total. The final calculation would be verified using the prespecified model and software, and would document all assumptions. The assumed effect and variability are planning values, not observed clinical evidence.

2.3 Primary endpoint model

The primary endpoint would be analysed using analysis of covariance (ANCOVA), with Week 24 HbA1c change as the dependent variable, treatment as a fixed effect, and baseline HbA1c and randomisation strata as covariates. The treatment contrast would estimate the adjusted mean difference between Study Drug X and placebo, with a 95% CI and p-value. Adjusting for baseline can improve precision when baseline measurements are associated with follow-up values. Model assumptions, including residual distribution and influential observations, would be examined using diagnostic plots and appropriate checks. If assumptions were materially violated, a prespecified robust or alternative analysis would be reported as a sensitivity analysis rather than chosen solely because it produced a preferred result.

2.4 Secondary endpoints and multiplicity

Continuous secondary outcomes, such as fasting plasma glucose and body weight, would be analysed using ANCOVA or an appropriate repeated-measures model, adjusted for baseline and relevant stratification factors. Binary outcomes, such as the proportion achieving HbA1c below 7.0%, would be summarised by group and compared using a suitable regression model or stratified method; absolute risk differences and confidence intervals would be presented where possible. Time-to-event methods would be used only for endpoints for which timing is meaningful and event counts support reliable estimation.

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2.5 Missing data and intercurrent events

The primary estimand would specify the population, treatment conditions, endpoint, summary measure, and handling of intercurrent events in line with ICH E9(R1) principles. The main analysis would use a prespecified approach for missing Week 24 data, such as multiple imputation under a missing-at-random assumption, incorporating baseline values, earlier post-baseline measurements, treatment group, and relevant prognostic variables. Imputed datasets would be analysed using the primary model and combined using Rubin’s rules.

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2.6 Safety analysis and data presentation

Safety outcomes would be summarised descriptively by treatment group. The report would include the number and percentage of participants experiencing any treatment-emergent adverse event, serious adverse events, events leading to discontinuation, deaths, and events of special interest. Exposure-adjusted rates could supplement participant-level proportions if treatment exposure differed materially between groups. Laboratory results would be summarised at baseline and follow-up, with shifts into clinically important categories described. The study would not be expected to have adequate power to detect rare harms; absence of a statistically significant difference would not establish equivalent safety.

2.7 Interim review, deviations, and transparency

Any interim analysis affecting treatment decisions would require prespecified timing, access controls, and appropriate adjustment for repeated testing if inferential conclusions were intended. For a small Phase II study without a planned efficacy interim analysis, routine blinded safety review and data-quality monitoring may be more appropriate. All analyses not specified before unblinding would be clearly labelled post hoc. Results would be presented with participant denominators, units, variability measures, CIs, and missing-data counts. The statistical report would distinguish prespecified analyses from exploratory analyses and disclose deviations from the SAP.

3. Discussion of Results

3.1 Summary of illustrative findings

For the purpose of demonstrating interpretation, assume that the hypothetical trial randomised 200 participants, with 100 assigned to each group. At Week 24, the adjusted mean HbA1c change is assumed to be −0.8 percentage points with Study Drug X and −0.3 percentage points with placebo, giving an adjusted between-group difference of −0.5 percentage points (95% CI −0.8 to −0.2; p = 0.002). These values are illustrative only and are not evidence about an actual medicine. In a real report, every number in this section must be replaced by results generated from the locked study database and final SAP.

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3.2 Clinical meaning and secondary outcomes

A reduction in HbA1c may support improved glycaemic control, but the relevance of a particular change depends on individual circumstances, current treatment options, and the balance between benefits and harms. The hypothetical primary result would be more persuasive if supported by consistent changes in fasting plasma glucose and by a reasonable proportion of participants reaching the prespecified HbA1c target. If secondary outcomes were inconsistent, that inconsistency would require explanation rather than selective emphasis. For example, a reduction in HbA1c without a corresponding change in fasting glucose might reflect differences in measurement timing, post-prandial effects, variability, or chance.

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3.3 Safety and benefit–risk interpretation

The hypothetical safety profile would need to be considered alongside efficacy. Similar overall adverse-event proportions between groups could be reassuring, but would not rule out uncommon, delayed, or subgroup-specific risks. Investigators should examine event type, severity, timing, recurrence, relationship to treatment, discontinuations, laboratory abnormalities, and exposure duration. An imbalance in a serious event—even if based on few participants—would warrant clinical review and possibly further investigation rather than dismissal on the basis of statistical significance alone.

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3.4 Strengths of the study design

Important strengths of the hypothetical design include random allocation, double blinding, a concurrent placebo control, stable background therapy, central laboratory testing, prespecified endpoints, and a documented SAP. Together, these features reduce several common sources of bias and support a clearer estimate of the treatment effect. Standardised visit schedules and structured safety collection also improve consistency across sites. A transparent participant flow diagram and clear reporting of deviations would help readers judge whether the planned design was implemented as intended.

3.5 Limitations and threats to interpretation

Several limitations would remain. First, the hypothetical sample size and 24-week follow-up would be insufficient to characterise rare harms or long-term clinical outcomes. Second, exclusion criteria may limit generalisability to older adults, people with substantial comorbidity, pregnant individuals, or patients with more advanced renal or hepatic impairment. Third, trial participation and frequent follow-up can produce adherence and monitoring levels that differ from routine care. Fourth, missing outcome data and treatment discontinuation may bias estimates if the assumptions used in the primary analysis are not credible.

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3.6 Implications and future research

If the illustrative efficacy signal were confirmed in actual data and the safety profile were acceptable, the results could justify a larger, longer, adequately powered trial. Future studies should confirm the effect in a more diverse population, include active comparators where clinically appropriate, and assess persistence of glycaemic control. Longer follow-up should characterise uncommon adverse events and clinically meaningful outcomes. Research should also evaluate patient-reported outcomes, treatment satisfaction, adherence, health-resource use, and whether benefits are maintained in routine clinical practice.

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A further consideration is the distinction between internal validity and external validity. Internal validity concerns whether the observed difference can reasonably be attributed to assigned treatment rather than bias, chance, or differential care. Randomisation, allocation concealment, blinding, consistent outcome measurement, and prespecified analysis primarily support internal validity. External validity concerns whether the result is applicable to patients and settings beyond the trial. It depends on how participants were recruited, the diversity of the enrolled population, background therapies, clinical monitoring, and how closely trial procedures resemble routine practice. A carefully controlled study can produce a reliable estimate for its enrolled population while still leaving uncertainty about effectiveness in everyday care. The final report should therefore describe baseline characteristics, recruitment settings, concomitant medicines, adherence, and reasons for withdrawal in enough detail for readers to judge applicability. It should avoid implying that an average treatment effect will be identical for every patient. Clinicians may need to consider age, comorbidities, renal function, concurrent medicines, patient preferences, and affordability when applying evidence. These considerations do not invalidate a randomised trial; they clarify the boundaries within which its conclusions are most defensible. A plain-language summary can communicate the main findings and uncertainties to participants and the public, while technical appendices can preserve detailed methods for specialist review.

3.7 Overall conclusion

The model illustrates how a clinical study report can connect design choices, statistical methods, and clinical interpretation. A credible report does not simply declare a treatment effective because a p-value is below 0.05. It explains who was studied, how bias and risk were managed, how missing data and multiple comparisons were handled, how precise the effect estimate is, and how limitations affect generalisability. For an actual study, conclusions must be based only on verified results and should be proportionate to the design, duration, and totality of evidence.

Selected References

1. International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH). E6(R3) Guideline for Good Clinical Practice. 2025.
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5. Schulz KF, Altman DG, Moher D; CONSORT Group. CONSORT 2010 Statement: updated guidelines for reporting parallel group randomised trials. BMJ. 2010;340:c332.

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7. National Research Council. The Prevention and Treatment of Missing Data in Clinical Trials. Washington, DC: National Academies Press; 2010.